2026 Science Nobel Prizes Go to Brain Light Switches, Ghost Particles and Mirror Molecules

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Every year, **Nobel Prizes** are given for some of the most important work in science. The 2026 prizes honored work in biology, physics and chemistry.

The physics prize went to the man behind the IceCube Neutrino Observatory at the South Pole. It helped find tiny, ghostly particles called **neutrinos** coming from outside our solar system. The chemistry prize honored work on **molecules** that are mirror images of each other, like your left and right hands.

The medicine prize went to Karl Deisseroth, Peter Hegemann and Georg Nagel. They found a way to turn brain cells on and off with light.

It started with a tiny green **alga**. Hegemann and Nagel found a **protein** in it that makes an electric signal when light hits it. Deisseroth's team put this **protein** into nerve cells. When light shined on them, the cells switched on. In mice, blue light even moved their whiskers.

Scientists hope this tool can help treat problems like blindness.

Word Treasure

Nobel Prizes -- Famous awards given each year for great work in science and other fields.
neutrinos -- Super tiny particles that fly through space and almost never hit anything.
molecules -- Tiny groups of atoms that make up everything around us.
alga -- A simple plant-like living thing that usually lives in water.
protein -- A tiny building block that does important jobs inside living things.

Background you need

The Nobel Prizes are some of the most famous awards in the world, and some are given each year for big science discoveries.

Your brain is full of nerve cells that talk to each other with tiny electric signals.

Being able to switch those cells on with light helps scientists learn what each part of the brain does.

Break it down

WHO: Karl Deisseroth, Peter Hegemann, Georg Nagel and the man behind IceCube
WHAT: They won 2026 Nobel Prizes for science discoveries
WHERE: The IceCube Neutrino Observatory at the South Pole and labs that study the brain
WHY: Their discoveries opened new ways to study the brain, space particles and molecules

Why it matters

These discoveries started with curious questions about tiny things, just like the questions kids ask every day.

Test what you remember

1. Which three kinds of science were honored by the 2026 prizes?

- (A) Cooking, sewing and gardening
- (B) Biology, physics and chemistry
- (C) Math, reading and spelling
- (D) Art, music and dance

2. Where is the IceCube Neutrino Observatory?

- (A) At the South Pole
- (B) Under the ocean
- (C) On the Moon
- (D) In a city park

3. What are neutrinos?

- (A) Small animals
- (B) Tiny, ghostly particles
- (C) Types of plants
- (D) Big rocks from space

4. The chemistry prize was about molecules that are like what?

- (A) Your left and right hands
- (B) Rainbows
- (C) Snowflakes
- (D) Square boxes

5. What did the medicine prize winners use to turn brain cells on and off?

- (A) Magnets
- (B) Light
- (C) Water
- (D) Sound

6. What did blue light make the mice do?

- (A) Change color
- (B) Run in circles
- (C) Move their whiskers
- (D) Sleep

Answer key (try the quiz first!): 1: B 2: A 3: B 4: A 5: B 6: C

Questions to think about

MY THOUGHTS (at least 20 words):